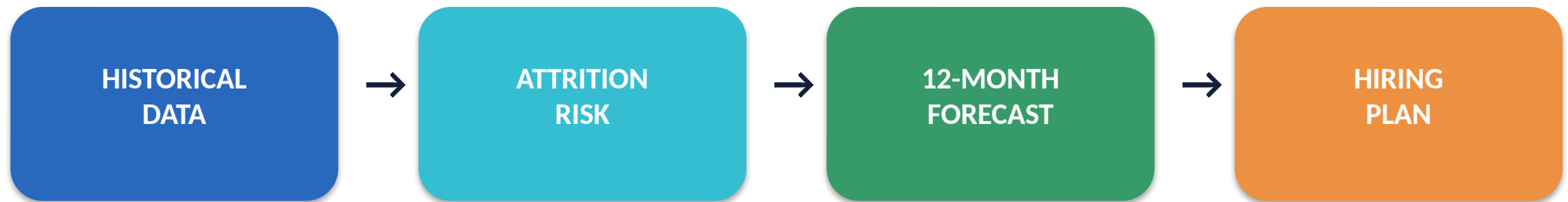


Workforce Planning & Forecasting

Predicting the right number of people — before the need becomes urgent.



Pavithra D • 2117240030101

Business Analytics Laboratory Mini Project

Why do we need forecasting?

GROWTH

More business → more employees needed

ATTRITION

Employees may leave unexpectedly

RISK

Under-hiring causes workload & shortages

DECISION

Forecasting supports proactive hiring

GOAL Forecast staffing needs for every department and role for the next 12 months.

From employee records to useful signals

~30,000

active employee records

6 departments

Engineering

Sales

Support

HR

Finance

Marketing

Tenure

How long has the employee worked?

Age

A risk factor in the scoring rule

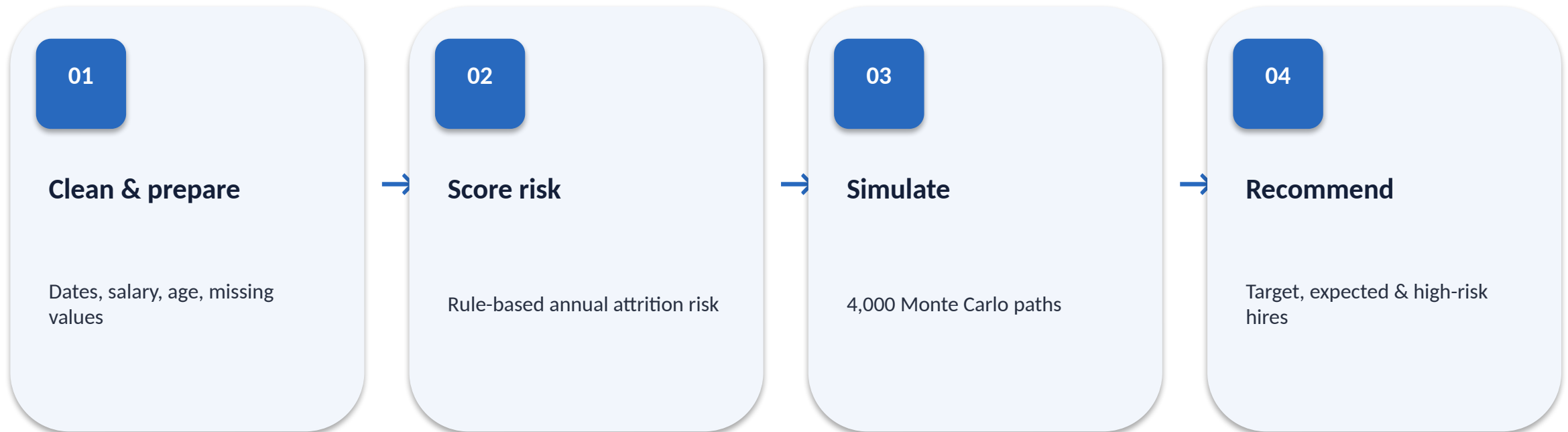
Salary

Compared with department Q1

Hiring Month

Build historical hiring trends

How the forecast works



Output: 12-month staffing forecast + P10/P90 uncertainty + hiring recommendations

Power BI Dashboard – Workforce Overview

1. Department Hiring Priorities



Figure 1. Horizontal bar chart of Sum of Recommended_Hires by Department, with KPI cards for Current, Target, Recommended, and High-Risk headcount totals.

This page ranks departments by hiring need. All six bars are visually near-identical in length (832), which is why the KPI cards on the right correctly total approximately 30K current headcount, 32K target headcount, and 5K in both recommended hires and high-risk hiring need — matching department-forecast table almost exactly.

Page 2 – Current vs Target Headcount



2. 12-Month Supply Trend

Page 4 – 12-Month Supply Trend and KPI Summary

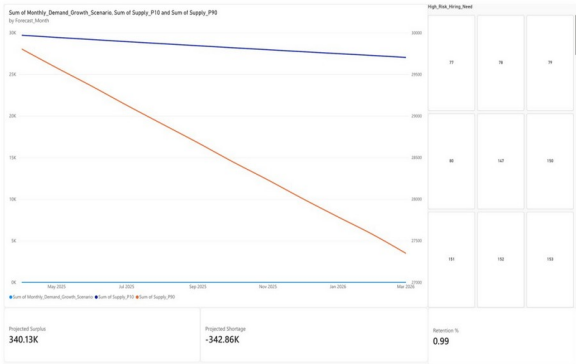


Figure 4. Line chart of Supply_P10 and Supply_P90 across the forecast horizon, with Projected Surplus, Projected Shortage, Retention %, and role-level High_Risk_Hiring_Need cards.

The declining Supply_P90 (optimistic survival) line and the comparatively flat, high Supply_P10 (pessimistic survival) line correctly reflect the model design: P90 represents the scenario where fewer employees exit than expected, so cumulative supply stays high, while P10 represents heavy-than-usual exits, causing supply to fall away over the 12-month horizon. The Projected Surplus (340.13K) and Projected Shortage (-342.86K) cards are large because they SUM Supply_P10/P90 headcount snapshot across all 6 departments and all 12 forecast months — s

Easy explanation:
The dashboard shows where hiring is needed and how staffing changes over 12 months.

Final Output & Key Result

Python / VS Code – Executive Staffing Priorities

```
force_forecast.py

WORKFORCE FORECAST COMPLETED SUCCESSFULLY

Department Current_Headcount Target_Headcount Expected_Exits Recommended_Hires High_Risk_Hiring_Need Annual_Attrition_Scenario Priority
Support      4990           5350           37.2           828           854           0.0978      High
HR            4976           5338           37.1           828           855           0.0979      High
Engineering   5070           5424           37.7           827           855           0.0976      High
Sales         5019           5374           37.3           824           850           0.0976      High
Finance       4938           5300           36.7           823           849           0.0976      High
Marketing     5007           5361           37.2           822           848           0.0976      High
PS C:\Users\pavit\AppData\Local\Programs\Microsoft VS Code>
```

How to explain

Current

Employees available now

Target

Employees needed after 12 months

Recommended

Normal hiring requirement

High-Risk

Extra buffer for higher attrition

Key result

822-828 recommended hires

848-855 high-risk need